

OVERVIEW

Have you ever wondered what to do when students solve a mathematical problem in different ways? Should you highlight only the most formal approach — or share multiple ways, and in what order? In this conversation, you'll consider the strengths of common mathematical representations, strategies for sequencing them, and the benefits of connecting them.

SPARK MEDIA & FOCUS LESSON

Student Work

Four students share different approaches to the same problem, in the lesson [Hair Today, Gone Tomorrow](#).

PEDAGOGICAL THEME

Deepening Mathematical Understanding

USING THE CONVO GUIDE FOR A GREAT ROUNDTABLE DISCUSSION

- This conversation is designed for a group of 2-5 educators over 30-60 minutes.
- The Convo Guide serves as an agenda for a roundtable conversation where everyone is there both to learn and to share their expertise.
- Each participant should have their own copy of the Convo Guide and access to the Spark Media.
- Pace yourselves to get through the entire agenda in the time you have.
- Questions aren't intended to have a single right answer; instead, they're designed to spark rich discussions.
- At the end, everyone will commit to trying "one small thing" the very next day. Consider how you might follow up with one another!

OBSERVE

5–15 MIN

- To start, take a look at the Spark Media.
- How are students solving the problem differently?
- What are some strengths and weaknesses in each representation?

DISCUSS

10–15 MIN

- If you were teaching this lesson, which approaches would you want to surface to the whole class for discussion? If you choose to surface multiple approaches, in what order would you want to present them?
- What connections would you want to help students make among the approaches — and how would you help them make those connections?

RELATE

10–15 MIN

- What does sequencing and sharing work look like in your classroom now? When you do this, what challenges do you experience?"
- Are there any shifts you'd like to make to the way you approach sequencing and sharing student work?
- To prepare to meaningfully sequence and share student work while *teaching*, what could you think about or do during your *lesson planning*?

...AND COMMIT!

5–10 MIN

- What is one small thing you can do tomorrow in support of those shifts?

SAMPLE STUDENT WORK

The following student work samples are related to the Citizen Math lesson Hair Today, Gone Tomorrow in which students use linear equations to explore the question: How long would it take to donate your hair to make a wig?

In place of these, you could use student work samples from one of your own classrooms. In all cases, you'll want everyone to look at the same work samples, and they should show different valid mathematical approaches for the same problem.

Q1. Each year, thousands of children lose their hair due to cancer or other medical conditions. This can be traumatic. Fortunately, there are organizations that use donated hair to create wigs for sick children.

Imagine the following three people want to donate their hair. For hair to be accepted, it must be at least 10 inches in length. If everyone's hair grows at a rate of 0.5 inches per month*, how many months will it be until each person is eligible to donate?

(*In reality, human hair grows at different rates around the world. According to a 2005 study from the International Journal of Dermatology, people from Africa had hair that grew at a rate of 0.33 inches per month. People from Asia had hair that grew at a rate of 0.53 inches per month. And people from other parts of the world had hair that grew at rates somewhere in between.)

STUDENT A



current length: 0 inches

month	length
0	0 in
1	0.5 in
2	1.0 in
3	1.5 in
4	2.0 in
5	2.5 in
10	5.0 in
20	10.0 in



current length: 8 inches

month	length
0	8.0 in
1	8.5 in
2	9.0 in
3	9.5 in
4	10.0 in



current length: 12.5 inches

month	length
0	12.5 in
-1	12.0 in
-2	11.5 in
-3	11.0 in
-4	10.5 in
-5	10.0 in

Q1. ...Imagine the following three people want to donate their hair. For hair to be accepted, it must be at least 10 inches in length. If everyone's hair grows at a rate of 0.5 inches per month*, how many months will it be until each person is eligible to donate?

STUDENT B

		
$10 - 0 = 10 \text{ inches}$	$10 - 8 = 2 \text{ inches}$	$10 - 12.5 = -2.5 \text{ inches}$
$10 = 0.5t$	$2 = 0.5t$	$-2.5 = 0.5t$
$t = 20 \text{ months}$	$t = 4 \text{ months}$	$t = -5 \text{ months}$

STUDENT C

		
$0 + 0.5m = 10$	$8 + 0.5m = 10$	$12.5 + 0.5m = 10$
$0.5m = 10$	$0.5m = 2$	$0.5m = -2.5$
$\frac{0.5}{0.5} \quad \frac{10}{0.5}$	$\frac{0.5}{0.5} \quad \frac{2}{0.5}$	$\frac{0.5}{0.5} \quad \frac{-2.5}{0.5}$
$m = 20$	$m = 4$	$m = -5$

Q1. ...Imagine the following three people want to donate their hair. For hair to be accepted, it must be at least 10 inches in length. If everyone's hair grows at a rate of 0.5 inches per month*, how many months will it be until each person is eligible to donate?

STUDENT B



current length: 0 inches

$$\text{slope} = 0.5 \text{ in/month}$$

$$y\text{-int} = 0 \text{ inches}$$



current length: 8 inches

$$\text{slope} = 0.5 \text{ in/mo.}$$

$$y\text{-int} = 8 \text{ inches}$$



current length: 12.5 inches

$$\text{slope} = 0.5 \text{ in/mo.}$$

$$y\text{-int} = 12.5 \text{ inches}$$

HAIR LENGTH & HAIRCUTS

